

DP014

Physics 1

Semester 1

Session 2024/ 2025



KOLEJ

MATRIKULASI

PAHANG

**MODEL PAPER
SEMESTER I**

Physics Unit

**JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.**

ARAHAN KEPADA CALON :

Kertas soalan ini mengandungi 7 soalan.
Jawab semua soalan.

INSTRUCTIONS TO CANDIDATE :

*This question paper consists of 7 questions.
Answer all questions.*

UNTUK KEGUNAAN PEMERIKSA		
NO. SOALAN	MARKAH PENUH	MARKAH DIPEROLEH
1	8	
2	12	
3	9	
4	12	
5	9	
6	12	
7	18	
JUMLAH	80	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$

Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19}\text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9\text{ N m}^2\text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5\text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000\text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|---|
| 1. $v = u + at$ | 23. $\theta = \omega_o t + \frac{1}{2} \alpha t^2$ |
| 2. $s = ut + \frac{1}{2} at^2$ | 24. $\omega^2 = \omega_o^2 + 2\alpha\theta$ |
| 3. $v^2 = u^2 + 2as$ | 25. $\theta = \frac{1}{2}(\omega_o + \omega)t$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 26. $\tau = rF \sin \theta$ |
| 5. $v = u - gt$ | 27. $\frac{Q}{t} = -kA \left(\frac{\Delta T}{L} \right)$ |
| 6. $v^2 = u^2 - 2gs$ | 28. $v_{rms} = \sqrt{\langle v^2 \rangle}$ |
| 7. $s = ut - \frac{1}{2} gt^2$ | 29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$ |
| 8. $p = mv$ | 30. $\Delta U = Q - W$ |
| 9. $J = F \Delta t$ | |
| 10. $J = \Delta p = mv - mu$ | |
| 11. $f_s \leq \mu_s N$ | |
| 12. $f_k = \mu_k N$ | |
| 13. $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$ | |
| 14. $K = \frac{1}{2} mv^2$ | |
| 15. $U = mgh$ | |
| 16. $U_s = \frac{1}{2} kx^2 = \frac{1}{2} Fx$ | |
| 17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$ | |
| 18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$ | |
| 19. $s = r\theta$ | |
| 20. $v = r\omega$ | |
| 21. $a_t = r\alpha$ | |
| 22. $\omega = \omega_o + \alpha t$ | |

SIRI 5 (PSPM SET 3)

- 1 (a) Convert physical quantities into SI unit.

- (i) 25 g cm^{-3}
- (ii) 8.5 km h^{-1}

[2 marks]

- (b) **FIGURE 1** below shows three coplanar vectors lying on the x-y plane act on a body. Determine S if $S = P + Q - R$.

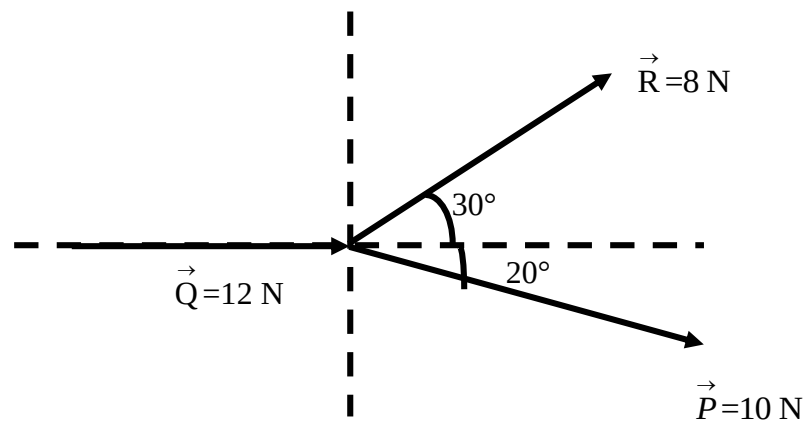


FIGURE 1

[6 marks]

- 2 (a) A train pulls out of a station and accelerates steadily for 20 s until its velocity reaches 8 m s^{-1} . It then travels at a constant velocity for 100 s, then it decelerates steadily for 30 s and finally stop.
- (i) Sketch a velocity-time graph for the journey.
 - (ii) Calculate the acceleration for each part of the journey **AND** the distance travelled for the whole journey.

[7 marks]

- (b) A taxi moving with an initial velocity 25 ms^{-1} is decelerated at a constant rate of 5 ms^{-2} in order to stop in front of a roadblock. Determine the stopping distance of the taxi.

[2 marks]

- (c) A ball is thrown upwards with a velocity of 3 m s^{-1} and it reaches its maximum height. Calculate the maximum height and the time taken to reach the maximum height.

[3 marks]

3 (a)

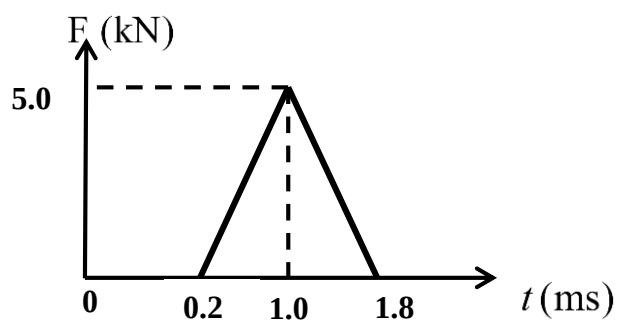


FIGURE 3.

FIGURE 3 shows a force-time graph for a tennis ball of mass 80 g. Calculate

- (i) the impulse of the force.
- (ii) the final velocity of the object if it is initially at rest.

[4 marks]

(b)

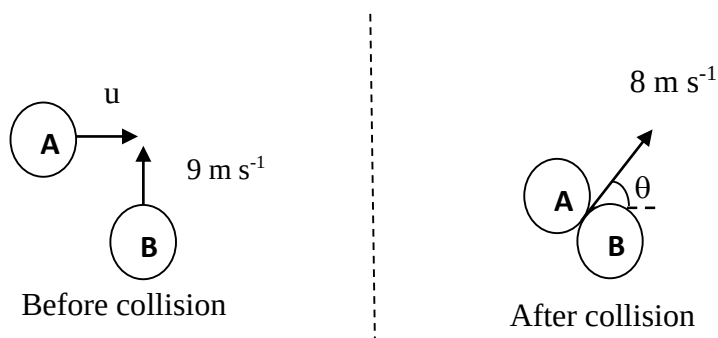


FIGURE 3.2

FIGURE 3.2 shows two objects with equal mass before and after they collide. After collision, they move together. Calculate the initial velocity of object A.

[5 marks]

- 4 (a) A body is acted by four forces as shown in the **FIGURE 4a** below.

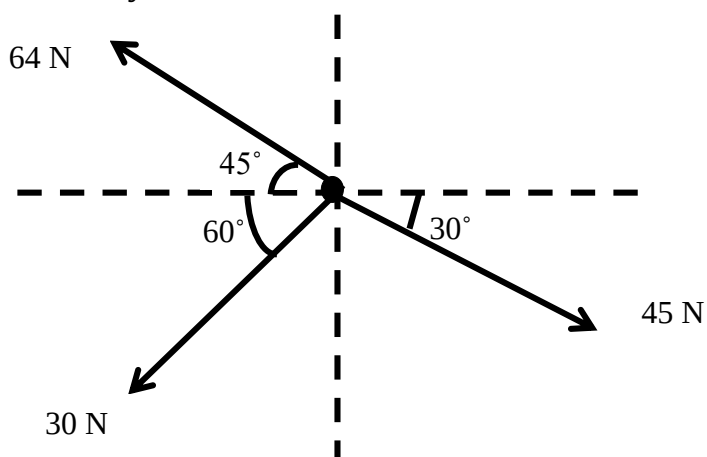


FIGURE 4a

If the body is in equilibrium, calculate the value of F .

[4 marks]

- (b) A block, P of mass 3 kg is placed on an inclined plank as shown in the **FIGURE 4b** below.

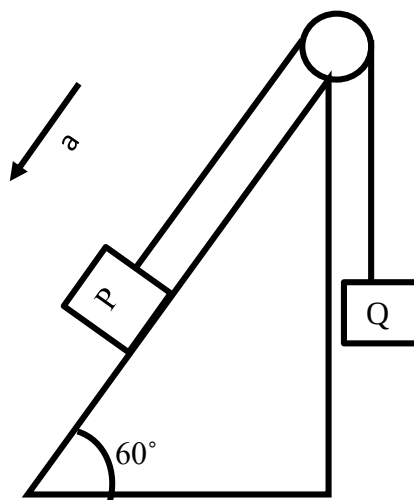


FIGURE 4b

The block P of a mass 3 kg is connected to another block is block, Q of mass 1 kg by a string through a smooth pulley.

- Sketch with label the free body diagrams which show all the forces that act on P and Q respectively.
- When block P is released from rest, it moves under a constant frictional force of 5 N . Find the acceleration of block P .

[8 marks]

5 (a)

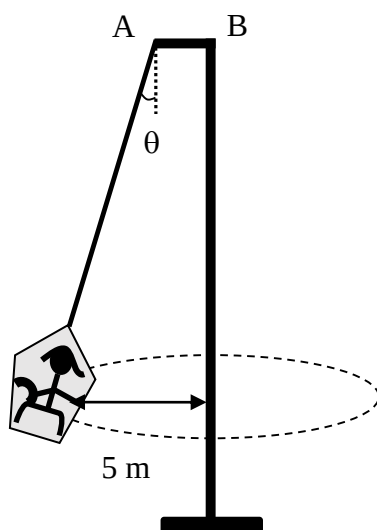


FIGURE 5

FIGURE 5 shows part of a fairground roundabout. AB is a horizontal arm which rotates about a fixed-point A. The chair which is attached to AB by a cable, moves in a horizontal circle of radius 5.0 m completing each revolution in 6.0 s. The weight of chair and occupant is 800 N.

- (i) Calculate the speed of the chair
- (ii) Calculate the force required to maintain the circular motion of the chair.
- (iii) Determine the angle, θ between the cable and vertical

[6 marks]

- (b) A toy of mass 100 g is attached to one string of 1.2 m long. The toy makes 2 rps in horizontal circle. Determine the centripetal acceleration of the toy.

[3 marks]

- 6 (a) A wheel rotating at 10.5 rad s^{-1} takes 5.2 s to stop. Calculate

- (i) the angular acceleration of the wheel.
- (ii) the number of revolutions the wheel makes before it stops.

[5 marks]

(b)

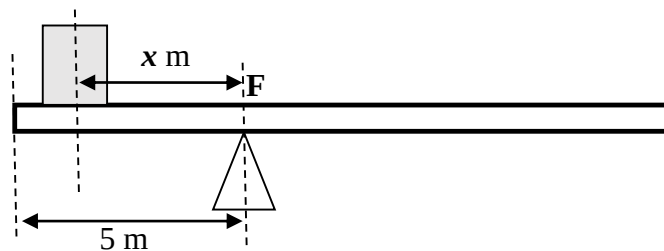


FIGURE 6

FIGURE 6 shows a 14.0 m long rigid uniform plank of mass 20 kg is supported with a fulcrum at point F. A box of mass 10 kg is placed on the plank at a distance x m from F. N_F is normal contact force between the plank and fulcrum F.

- (i) Sketch all forces acting on the plank.
- (ii) Calculate N_F and x when resultant torque at point F is 0 Nm.

[7 marks]

- 7 (a) (i) An aluminum rod has a diameter of 3 cm and thickness of 0.6 m. One end of the rod is placed in the boiling water and the other end is placed in ice. Calculate the quantity of heat transferred through the rod within 1 minute.
Given, $k = 205 \text{ Wm}^{-1}\text{K}^{-1}$

- (ii) Find the difference temperature in temperature across surfaces of an iron plate that is 5 cm thick, through every 100 m^2 of which heat flows at the rate of $96 \times 10^4 \text{ J}$ per second. [$k = 96 \text{ Js}^{-1}\text{m}^{-1}\text{K}^{-1}$]

[4 marks]

- (b) (i) A 515 cm^3 flask contains 0.460 g of a gas at a pressure of 153 kPa and a temperature of 322 K. What is the molecular mass of this gas?

- (ii) What is the temperature of a gas of CO_2 molecules whose rms speed is 339 ms^{-1} ?
(Given the molar mass of CO_2 is 44 g/mol)

- (iii) The molar mass of oxygen is 32 g mol^{-1} at 300 K, calculate the root mean square speed of the oxygen molecule.
[$k = 1.38 \times 10^{-23} \text{ J s}^{-1} \text{ m}^{-1} \text{ K}^{-1}$]

[7 marks]

- (c) (i) If 100 J of heat added to a system that does 40 J of external work, by how much is the internal energy of that system raised?
- (ii) Sketch on the same axis p - V graph for each of the following process:
- Isothermal
 - Isobaric
 - Adiabatic
- (iii) An ideal gas trapped in a cylinder with a frictionless movable piston is heated by supplying 100 J of heat to it. It expands from a volume of 300 cm^3 to 500 cm^3 at a constant pressure of $1 \times 10^5 \text{ Pa}$. What is the change in the internal energy of the gas?

[7 marks]

END OF QUESTION PAPER